STAFF REVIEW · H446 094 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Requested item found in cache.

2. A unit of processor clock timing.

3. An independent CPU execution unit.

4. 64→32→16→8→4→2→1; six halvings.

I Do — source worked steps

1. Explain the mechanism — 3 GHz offers 50% more cycles each second than2 GHz; this does not universally imply 50% faster program runtime. — The comparison requires equal microarchitecture and workload assumptions.

We Do — guided CHECK

1. Explain the mechanism
It may not be substantially faster from extra cores alone; only one thread is the bottleneck.
Parallel benefits need tasks/threads that can run concurrently.

You Do — source mark schemes

1. Clock affects cycles/sec [2], cores parallel tasks [2], cache reduces slower memory accesses [2].

2. 3/2=1.5 [2], Y50% more cycles/s [2], memory stalls/parallelisation or instruction mix [2].

3. Rendering often exploits cores [2], sequential script needs stronger single-thread performance [2], cache/IPC/workload caveats [2].

Exit Ticket — source answer

1. False. Clock speed alone does not determine total performance.